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Topic: Pandas

```
import pandas as pd
from google.colab import drive
```

```
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
file_path = '/content/drive/MyDrive/Colab_Notebooks/data_Iris.csv'

data = pd.read_csv(file_path, delimiter=',', )
print(data.head())
```

```
   Sepal.Length  Sepal.Width  Petal.Length  Petal.Width Species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

```
print(data.tail())
```

```
     Sepal.Length  Sepal.Width  Petal.Length  Petal.Width    Species
145           6.7          3.0           5.2          2.3  virginica
146           6.3          2.5           5.0          1.9  virginica
147           6.5          3.0           5.2          2.0  virginica
148           6.2          3.4           5.4          2.3  virginica
149           5.9          3.0           5.1          1.8  virginica
```

```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Sepal.Length  150 non-null    float64
 1   Sepal.Width   150 non-null    float64
 2   Petal.Length  150 non-null    float64
 3   Petal.Width   150 non-null    float64
 4   Species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

```
data.shape
```

```
(150, 5)
```

```
print(data.describe())
```

```
       Sepal.Length  Sepal.Width  Petal.Length  Petal.Width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198667
std        0.828066     0.433594      1.764420     0.763161
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
```

```
data.columns
```

```
Index(['Sepal.Length', 'Sepal.Width', 'Petal.Length', 'Petal.Width',
       'Species'],
      dtype='object')
```

```
data['Species'].head()
```

	Species
0	setosa
1	setosa
2	setosa
3	setosa
4	setosa

dtype: object

```
data.loc[0:5]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa

```
data.iloc[0:5]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
data.loc[0:"Sepal.Length"]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

```
data[data['Species'] == "setosa"].head()
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
data[data["Sepal.Length"] > 5].head()
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
10	5.4	3.7	1.5	0.2	setosa
14	5.8	4.0	1.2	0.2	setosa
15	5.7	4.4	1.5	0.4	setosa

```
data["sepal_area"] = data["Sepal.Length"] * data["Sepal.Width"]
data
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	sepal_area
0	5.1	3.5	1.4	0.2	setosa	17.85
1	4.9	3.0	1.4	0.2	setosa	14.70
2	4.7	3.2	1.3	0.2	setosa	15.04
3	4.6	3.1	1.5	0.2	setosa	14.26
4	5.0	3.6	1.4	0.2	setosa	18.00
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica	20.10
146	6.3	2.5	5.0	1.9	virginica	15.75
147	6.5	3.0	5.2	2.0	virginica	19.50
148	6.2	3.4	5.4	2.3	virginica	21.08
149	5.9	3.0	5.1	1.8	virginica	17.70

150 rows × 6 columns

Next steps: Generate code with data New interactive sheet

```
data.drop("sepal_area", axis=1, inplace=True)
```

```
data
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

Next steps: Generate code with data New interactive sheet

```
data.dropna() # removes missing values
data.dropna(subset=["Sepal.Length"] ) # it will remove that row with missing Speal.Length is removed.
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

```
data.isnull().sum()
```

	0
Sepal.Length	0
Sepal.Width	0
Petal.Length	0
Petal.Width	0
Species	0

dtype: int64

```
data.sort_values("Sepal.Length")
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
13	4.3	3.0	1.1	0.1	setosa
8	4.4	2.9	1.4	0.2	setosa
42	4.4	3.2	1.3	0.2	setosa
38	4.4	3.0	1.3	0.2	setosa
41	4.5	2.3	1.3	0.3	setosa
...	...	...	...	...	...
122	7.7	2.8	6.7	2.0	virginica
117	7.7	3.8	6.7	2.2	virginica
118	7.7	2.6	6.9	2.3	virginica
135	7.7	3.0	6.1	2.3	virginica
131	7.9	3.8	6.4	2.0	virginica

150 rows × 5 columns

```
data.sort_values("Sepal.Length", ascending=False)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
131	7.9	3.8	6.4	2.0	virginica
122	7.7	2.8	6.7	2.0	virginica
118	7.7	2.6	6.9	2.3	virginica
117	7.7	3.8	6.7	2.2	virginica
135	7.7	3.0	6.1	2.3	virginica
...	...	...	...	...	...
41	4.5	2.3	1.3	0.3	setosa
42	4.4	3.2	1.3	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
38	4.4	3.0	1.3	0.2	setosa
13	4.3	3.0	1.1	0.1	setosa

150 rows × 5 columns

```
# loc[row, column]
data.loc[0, "Sepal.Length"] = 100
data.head()
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	100.0	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with data New interactive sheet

```
data[data["Sepal.Length"] == 100]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	100.0	3.5	1.4	0.2	setosa

```
data.loc[0:4]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	100.0	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
a=data.loc[[0,0,1,1]]
a.duplicated()
```

	0
0	False
0	True
1	False
1	True

dtype: bool

```
data.duplicated().value_counts()
```

	count
False	147
True	3

dtype: int64

```
# To show the duplicated value. note previous row is same as next row
data[data.duplicated()]
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
34	4.9	3.1	1.5	0.1	setosa
37	4.9	3.1	1.5	0.1	setosa
142	5.8	2.7	5.1	1.9	virginica

```
# To remove the duplicated values.
data.drop_duplicates().head(10)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	100.0	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
5	5.4	3.9	1.7	0.4	setosa
6	4.6	3.4	1.4	0.3	setosa
7	5.0	3.4	1.5	0.2	setosa
8	4.4	2.9	1.4	0.2	setosa
9	4.9	3.1	1.5	0.1	setosa

```
# To check how many rows are removed.
print("Before:",data.shape)
data = data.drop_duplicates()
print("After:",data.shape)
```

```
Before: (150, 5)
After: (147, 5)
```

Practice

1. Show first 10 rows
2. Show petal_length column
3. Filter rows where species = "versicolor"
4. Create Petal_area column
5. Sort by petal_width descending
6. Show average sepal_length

```
data.head(5)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
0	100.0	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with data | New interactive sheet

```
data["Petal.Length"].head()
```

	Petal.Length
0	1.4
1	1.4
2	1.3
3	1.5
4	1.4

dtype: float64

```
data[data['Species'] == "versicolor"].head()
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
50	7.0	3.2	4.7	1.4	versicolor
51	6.4	3.2	4.5	1.5	versicolor
52	6.9	3.1	4.9	1.5	versicolor
53	5.5	2.3	4.0	1.3	versicolor
54	6.5	2.8	4.6	1.5	versicolor

```
data["Petal.Area"] = data["Petal.Length"] * data["Petal.Width"]
data.head()
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	Petal.Area
0	100.0	3.5	1.4	0.2	setosa	0.28
1	4.9	3.0	1.4	0.2	setosa	0.28
2	4.7	3.2	1.3	0.2	setosa	0.26
3	4.6	3.1	1.5	0.2	setosa	0.30
4	5.0	3.6	1.4	0.2	setosa	0.28

Next steps: Generate code with data | New interactive sheet

```
data.sort_values("Petal.Width", ascending=False)
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species	Petal.Area
100	6.3	3.3	6.0	2.5	virginica	15.00
144	6.7	3.3	5.7	2.5	virginica	14.25
109	7.2	3.6	6.1	2.5	virginica	15.25
114	5.8	2.8	5.1	2.4	virginica	12.24
140	6.7	3.1	5.6	2.4	virginica	13.44
...	...	...	...	...	...	...
39	5.1	3.4	1.5	0.2	setosa	0.30
12	4.8	3.0	1.4	0.1	setosa	0.14
13	4.3	3.0	1.1	0.1	setosa	0.11
9	4.9	3.1	1.5	0.1	setosa	0.15
32	5.2	4.1	1.5	0.1	setosa	0.15

147 rows × 6 columns

```
data['Sepal.Length'].mean()
```

```
np.float64(6.50204081632653)
```

```
data.groupby("Species").mean()
```

```
Sepal.Length  Sepal.Width  Petal.Length  Petal.Width  Petal.Area
```

```
# Aggregation functions are functions that combine multiple values into one single value.
# So one group -> multiple summary results.
data.groupby("Species").agg(["max", "min", "mean"])
```

	Sepal.Length			Sepal.Width			Petal.Length			Petal.Width			Petal.Area		
	max	min	mean	max	min	mean	max	min	mean	max	min	mean	max	min	mean
Species															
setosa	100.0	4.3	6.987500	4.4	2.3	3.431250	1.9	1.0	1.462500	0.6	0.1	0.250000	0.96	0.11	0.371667
versicolor	7.0	4.9	5.936000	3.4	2.0	2.770000	5.1	3.0	4.260000	1.8	1.0	1.326000	8.64	3.30	5.720400
virginica	7.9	4.9	6.604082	3.8	2.2	2.979592	6.9	4.5	5.561224	2.5	1.4	2.028571	15.87	7.50	11.328980

```
data["Species"].value_counts()
```

	count
Species	
versicolor	50
virginica	49
setosa	48

dtype: int64

Pivot Tables

For summaring data, short summary table

```
# pd.pivot_table(data, values="Sepal.Length", index="Species", aggfunc="mean")
data.pivot_table(values="Sepal.Length", index="Species", aggfunc="mean")
```

	Sepal.Length
Species	
setosa	6.987500
versicolor	5.936000
virginica	6.604082

```
# values->calculate value
data.pivot_table(values="Sepal.Length", index="Species", aggfunc=["mean", "nunique"])
```

	mean	nunique